

Modeling Residential Loss Severity

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1 The Components of Loss Severity

Conceptually, we can break down all the components of loss severity as follows:

- **Base Severity:** The difference between the original loan amount and the sale price of the house, expressed as a fraction of the principal balance.
- **Real Estate Commissions:** Amount due to the real estate broker upon the sale of the house plus any seller-paid closing costs. The broker amount usually averages 6%.
- **Foreclosure Costs:** Legal expenses, delinquent portions of property and insurance tax payments.
- **Holding Costs:** Property tax, property insurance, utilities.
- **Uncollected interest:** Missed interest payments by the borrower.
- **Insurance payments:** Recoveries from mortgage insurance.

The above set of categories can be further collapsed into four overall groups:

1. **Base Severity**
2. **Uncollected interest**
3. **Average levels for various expenses**
4. **Insurance payments**

The next section discusses each of these components in turn.

2 Modeling the Components of Loss Severity

2.1 Base Severity

Since base severity is the largest component of loss severity, calculating base severity is clearly a crucial part of the loss severity calculation.¹ In theory, we could use zip-code level or MSA-level HPIs to estimate the value of the foreclosed home. However, any HPI we choose will give us home price appreciation for the *average*

¹According to the Wilson study ([Wilson95]), on average, approximately 41% of the total loss severity can be attributed to base severity.

What does
Freddie
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house, and not necessarily the appreciation experienced by a foreclosed one. Clearly, foreclosed properties are likely to experience less home price appreciation than the average property. Therefore, in addition to estimating the current equity position of the borrower using the appropriate HPI, we should also consider the volatility of home price growth in that area. Specifically, we can construct a “z-score” for the property underlying the loan as follows:

$$z(t) = \frac{EQ(t)}{\sigma(t)} \quad (1)$$

Here, $EQ(t)$ measures the current equity position of the borrower and is given by $1 - UPB(t)/H(t)$, where $H(t)$ is the property price at time t and $UPB(t)$ is the current unpaid principal balance of all outstanding mortgages on the property. $\sigma(t)$ is the standard deviation of HPI growth rates for properties in the region that the property is located after t quarters. This is the square root of $\alpha t + \beta t^2$, where α and β are two volatility parameters for each HPI series published by FHFA.²

By construction, high z -scores should be associated with lower loss severities. In addition, computing $z(t)$ as above allows us to control for a couple of factors. First, given two foreclosed properties with the same value of $EQ(t)$ but that are located in areas with different volatilities, $z(t)$ assigns a lower value to the property that’s in a higher volatility area. Next, the denominator $\sigma(t)$ tends to grow over time, capturing the fact that home price variability tends to grow over time. However, we have to limit the value of the age variable t to avoid volatility eventually decreasing over time since α is positive and β is negative. This can be achieved by capping t at $\frac{\alpha}{2\beta}$.

Another nuance to consider is that the price of the REO relative to the average property may be a function of market conditions. In fact, [Wilson95] appears to suggest that in a good market, REO properties may trade at a greater haircut to the average property than in a soft market. The argument being that a greater degree of adverse selection must be present for a property to default in a good market. More generally, while our initial formulation should capture the impact of housing market conditions on loss severity, there should also be a haircut term h that depends on the following list of factors:

- **Loan age at foreclosure.** Early stage defaults may be associated with fraud and thus lead to greater loss severities.
- **Legal issues.** Foreclosure laws prevailing in the State of default: Judicial versus Power-of-Sale versus equitable right of redemption.
- **Mortgage characteristics.** SATO, Loan Size. Higher loan sizes are empirically seen to be associated with lower loss severities since the end investor has a greater stake in resolving the situation because of the larger absolute dollar of loss to consider.
- **Loan purpose.** Greater risk of appraisal bias with refinance transactions.
- **Occupancy status.** Investor properties would be associated with higher loss severities since it is easier for investors to “hand over the keys.”
- **Property type.** Residence versus second home/vacation home. Primary residences may be more liquid and the owner may think of their second home/vacation home from the perspective of an investor.
- **Last payment status.** Pre-foreclosure sale/short sale/foreclosure sale/REO sale.

The importance of the **Last payment status** is worth emphasizing. Loans enter REO only if the house has failed to sell during foreclosure; therefore only the lowest house value houses tend to enter the REO. Also, REO has higher associated administrative costs as well as larger amounts of lost interest since the loan has not been making payments for a longer time.

In general, h will have to be estimated through a multivariable regression.

²FHFA currently releases volatility numbers at the state level but not at the MSA level.

2.2 Uncollected Interest

Missed interest payments by the borrower. This is approximated by:

$$\text{Lost Interest} = \frac{\text{Coupon}}{12} \times \text{Months of Lost Interest},$$

where Coupon is the yearly interest rate for the loan.

2.3 Expenses

There are three elements here: real estate commissions, foreclosure expenses, and holding costs. Real estate commissions show some dependence by state. Foreclosure expenses will also definitely vary by state because of the difference in property laws by state. Holding costs will depend upon time spent between foreclosure and REO. Once again, since this category includes property taxes, there may be some dependence by state since higher cost states have higher property taxes.

2.4 Recoveries from MI

In general, a PMI policy covers only a specified portion of the loss. This percentage is typically between 12% to 35% of the loss claim. Most lenders use the loss coverage scheme developed by Fannie Mae and Freddie Mac. For example, the standard coverage on a 25- to 30-year loan is as follows:

- 35% for LTVs between 95.01% and 100%;
- 30% for LTVs between 90.01% and 95%;
- 25% for LTVs between 85.01% and 90%; and
- 12% for LTVs that are 85% and under.

Of course, the insurance coverage will be different for FHA and VA loans and needs to be modeled as such. In practice, the **claim amount** on a defaulted loan is the coverage percentage applied to the sum of the remaining principal balance, the accrued interest, and the related foreclosure expenses.

3 Framework for Loss Severity Model

Loss Severity (LS) is usually expressed in terms of the unpaid principal balance of the loan (or sometimes the original principal balance). For forecasting purposes, we usually prefer expressing it as a function of the UPB of the mortgage loan at the time of default. Applying loss severity to the defaulted balance gives us the loss experienced on the loan:

$$LS = \frac{\text{Loss}}{\text{Unpaid Principal Balance}} \times 100 \quad (2)$$

For the model, we break the loss down into multiple terms corresponding to the overall categories we discussed above, with each term expressed as a function of the unpaid principal balance:

$$LS = LS_{\text{BaseSeverity}} + LS_{\text{UncollectedInterest}} + LS_{\text{Expenses}} - LS_{\text{InsuranceRecoveries}} \quad (3)$$

If the model produces a value less than zero, we set the loss severity to zero.

The sub-model for Base Severity can be expressed as below, with the term in the square brackets playing the role of the haircut, h , that was discussed above. We have listed only a few of the factors that can affect h , an exhaustive list can be found in our previous discussion. The variables denoted with a lower case “p” correspond to fitting parameters.

$$LS_{\text{BaseSeverity}} = [p_{\text{Const}} + p_{\text{Occ}} + p_{\text{Purch}} + \dots] \times z(t) \quad (4)$$

4 Integration with the Roll-Rate Model (Accordion)

Integrating the Loss Severity model with the Roll-Rate model takes some work because while a number of the variables (for example, months of lost interest) are easy enough to obtain from the data set, it is not so straightforward to estimate them when running a roll-rate model.

5 Other Ideas

- Cash flow timing for loss severity amounts: apply short-term rates in IR model for PVing
- Two time elements to model: time between default and foreclosure and time between foreclosure and property disposition
- Foreclosure expenses should vary by bankruptcy status of the borrower
- Servicer effects could potentially be included in the haircut calculation

References

- [1] Wilson, Donald. December 1995. “Residential Loss Severity in California: 1992 to 1995,” *Journal of Fixed Income*.